Examination Paper

Examination Session:	Year:	Exam Code:
May/June	2023	ENGI3491-WE01

Title: (BEng/MEng)

Digital Electronics and Digital Signal Processing

Time Allowed:	2 hours
Additional Material provided:	No
Reporting problems during the exam:	Queries about the instructions for this exam (rubric) should be directed to: eng.boe@durham.ac.uk. If you experience any difficulties uploading your exam answers to Gradescope report these immediately to: eng.boe@durham.ac.uk.

Instructions to Candidates:	This is a synchronous, time restricted exam. All submissions must be uploaded to Gradescope as per the published timetable. Answer ALL questions. Please answer each question on a new page. All workings must be shown. This is an online open-book exam. • *The examination attempt must be all your own work. You must not interact with another person (either one-to-one, by telephone, on the internet, in chat rooms, on social media, via smart watches, etc.), or use cheat sites, etc. whilst completing the exam. You must not discuss any examination papers with friends, relatives, current or past students of this or any other University before the end of the exam submission window.* • *Candidates are expected to answer all questions in their own words. Should close paraphrasing, direct quotation be necessary candidates must acknowledge the source of any material used in their answers (this includes text, images, diagrams, charts, tables and/or graphs). Full referencing is not expected, but a brief indication of the source of the material (e.g. name of author) must be given.*

Revision:	

Question 1

a) In your own words, describe the role of hardware description languages (HDL) in digital electronics design. Your answer should include a detailed discussion of the functionality that HDLs include to represent digital circuits including concurrency, behavioural and structural descriptions as well as a description of how they form part of programmable logic development process. The development process for FPGAs should also be outlined.

[40%]

b) An engineer has observed a failure of their FPGA circuit design. They decide to add a synchroniser circuit to resolve the problem as it appears to be metastability related. The precise specification of their system is not relevant but it has a CLK input and an asynchronous external input, Z.

i) Briefly explain what is meant by metastability and the role of a synchroniser circuit in resolving this.

ii) Write a VHDL code using port map which could be used to implement a synchroniser circuit to interface with the engineer's design. Sketch the circuit schematic that you code will produce. State any assumptions and use std_logic.

iii) To further aid the system testing, the engineer decides that they want to include a digital circuit which keeps count of the number of rising edges on the asynchronous input. It has one output, R, which becomes high when 5 or more clock edges have been detected. Write a VHDL process to fulfil this specification, noting that your code should not increase the likelihood of metastability failure. State any assumptions and use std_logic. You can refer to any code already written for part (ii).

[60%]

continued

Question 2

a) In your own words, describe how and why digital logic circuits are tested after manufacture. Your answer should include a discussion of why this is often a challenging stage, including the role of test vectors as well as the JTAG interface.

[40%]

b) An asynchronous sequential logic network-based controller is required for a coffee machine which accepts coin payment. The cost of a cup of coffee is £1.50 and the machine accepts combinations of 50p and £1 coins only. It only accepts one coin at a time and will dispense when £1.50 is paid, or exceeded. If the amount is exceeded, it does not give change and resets the payment count for the next customer.

i) Design an asynchronous sequential logic network which meets this specification. You should show all working, clearly describing each step. Finally, you should produce a complete set of simplified Boolean equations which could be used to implement your design in NAND gate logic only but you do not need to draw a circuit diagram.
Clearly state any assumptions.

ii) The coffee machine manufacturer implements your design using NAND gate logic with a worst-case propagation delay of 5 ns. They are concerned that someone might be able to overcome the system by inserting coins at a high frequency (although, they also note that this is likely to lead to overpayment by the customer!).

Estimate the maximum frequency for which your system could accept coins and comment on the likelihood of the system failing, in the way that the manufacture suggests, in a real world use case. Explain your answer and show all working.

[60%]

END